

No.	Solution	Remarks
1(a)	<u>Speed is a physical quantity and can only be defined in terms of other physical quantities.</u> Distance is a physical quantity, but <u>second is a unit for time</u> which cannot be used to define speed. (A more appropriate definition for speed is distance travelled per unit time.) <u>Common mistakes:</u> - speed is total distance over total time - speed is rate of change of distance per unit time - the speed must be resolved in horizontal and vertical components. (speed is not a vector. No resolving required.)	[1] [1]
1(b)	Time to reach maximum height: $t = \frac{u \sin \theta}{g}$ $R = (u \cos \theta)(2t)$ $= (u \cos \theta)\left(2 \frac{u \sin \theta}{g}\right)$ $= \frac{2u^2 \sin \theta \cos \theta}{g}$	[1] [1]
1(c)	Maximum R occurs when $\sin 2\theta = 1$. Hence $\theta = 45^\circ$	[1]
1(d)	$R_0 = \frac{u^2}{g}$ $g = \frac{u^2}{R_0}$ $= \frac{(12.6)^2}{16.3}$ $= 9.7399 \text{ m s}^{-2}$ $\frac{\Delta g}{g} = 2 \frac{\Delta u}{u} + \frac{\Delta R_0}{R_0}$ $\frac{\Delta g}{9.7399} = \frac{2(3\%) + 4\%}{100\%}$ $\Delta g = 0.97$ $= 1 \text{ m s}^{-2}$ Therefore, $g = (10 \pm 1) \text{ m s}^{-2}$	[1] for calculating g [1] for calculating Δg [1]

	<u>Common mistakes:</u> - $g = (10.0 \pm 1.0) \text{ m s}^{-2}$	
Q6 (V)	Magnitude of the error in QD5	[4] for comment